Volume

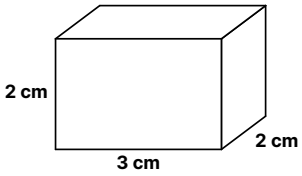
Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS volume *(rōrahi)* · cube *(mataono)* · rectangular prism *(poro tapawhā hāngai)* · layer · length *(roa)*

A Counting layers ●○○

Remember: Volume is the amount of space inside a solid. A cube with edges of 1 cm has a volume of 1 cubic centimetre (1 cm^3). Count the cubes in one layer, then multiply by the number of layers.

1. This box is 3 cm long, 2 cm wide and 2 cm high. It is filled with cubes that have edges of 1 cm.



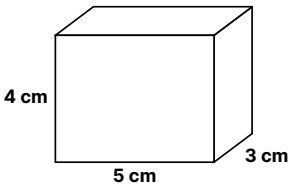
a) How many cubes fit in the bottom layer?

b) How many layers are needed to fill the box?

c) How many cubes fill the whole box?

d) What is the volume of the box?

2. This box is 5 cm long, 3 cm wide and 4 cm high. It is filled with cubes that have edges of 1 cm.



a) How many cubes fit in the bottom layer?

b) How many layers are needed to fill the box?

c) How many cubes fill the whole box?

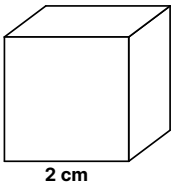
d) What is the volume of the box?

B Cubes ●●○

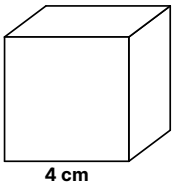
Remember: Every edge of a cube is the same length, so its volume is edge × edge × edge.

3. Find the volume of each cube.

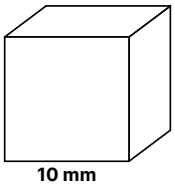
a)



b)



c)



4. Find the volume of:

a) A cube with edges of 3 m

b) A cube with edges of 5 cm

c) A cube with edges of 1 m

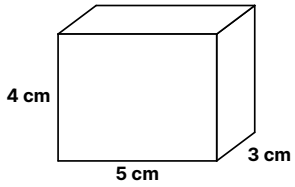
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C Rectangular prisms ●●○

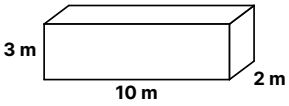
Remember: Volume of a rectangular prism = length × width × height.

5. Find the volume of each rectangular prism.

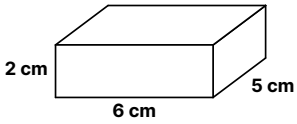
a)



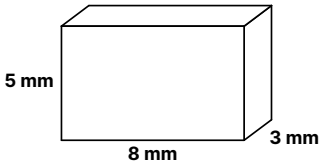
b)



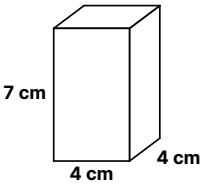
c)



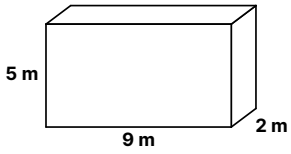
d)



e)



f)



6. Find the volume of:

a) A rectangular prism 12 cm by 5 cm by 2 cm

b) A rectangular prism 20 cm by 10 cm by 5 cm

D Units for length, area and volume ●●○

Remember: A length is measured in units (cm). An area is measured in square units (cm²). A volume is measured in cubic units (cm³).

7. Which unit would you use for each of these: cm, cm² or cm³?

a) The height of a cup _____

b) The amount of space inside a box _____

c) The size of a page in a book _____

d) The distance around a plate _____

e) The amount of wrapping paper that covers a present _____

f) The amount of sand in a sandpit _____

E Problems ●●●

8. A fish tank is 50 cm long, 30 cm wide and 20 cm deep. Find its volume.

9. A box is 4 cm by 3 cm by 2 cm. Find the total volume of 5 of these boxes.

10. A rectangular prism has a volume of 120 cm³. It is 6 cm long and 5 cm wide. Find its height.

11. A cube has a volume of 27 cm³. Find the length of its edges.

★ Extension

12. How many cubes with edges of 2 cm fit exactly into a box that is 6 cm by 4 cm by 2 cm?

13. 24 cubes with edges of 1 cm are used to build a rectangular prism. No edge of the prism is 1 cm long. Write the length, width and height of a prism that could be built.

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Volume

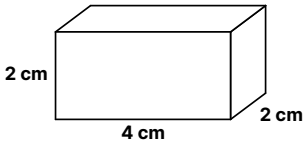
Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS volume *(rōrahi)* · cube *(mataono)* · rectangular prism *(poro tapawhā hāngai)* · layer · length *(roa)*

A Counting layers ●○○

Remember: Volume is the amount of space inside a solid. A cube with edges of 1 cm has a volume of 1 cubic centimetre (1 cm^3). Count the cubes in one layer, then multiply by the number of layers.

1. This box is 4 cm long, 2 cm wide and 2 cm high. It is filled with cubes that have edges of 1 cm.



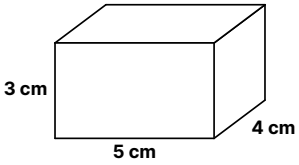
a) How many cubes fit in the bottom layer?

b) How many layers are needed to fill the box?

c) How many cubes fill the whole box?

d) What is the volume of the box?

2. This box is 5 cm long, 4 cm wide and 3 cm high. It is filled with cubes that have edges of 1 cm.



a) How many cubes fit in the bottom layer?

b) How many layers are needed to fill the box?

c) How many cubes fill the whole box?

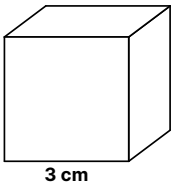
d) What is the volume of the box?

B Cubes ●●○

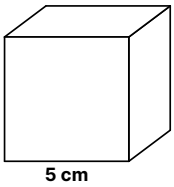
Remember: Every edge of a cube is the same length, so its volume is edge × edge × edge.

3. Find the volume of each cube.

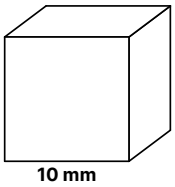
a)



b)



c)



4. Find the volume of:

a) A cube with edges of 2 m

b) A cube with edges of 4 cm

c) A cube with edges of 1 m

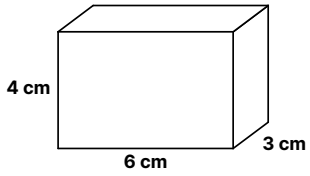
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C Rectangular prisms ●●○

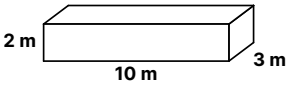
Remember: Volume of a rectangular prism = length × width × height.

5. Find the volume of each rectangular prism.

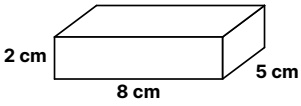
a)



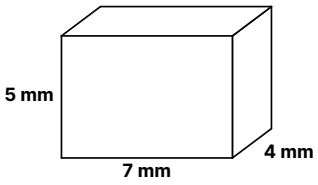
b)



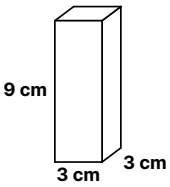
c)



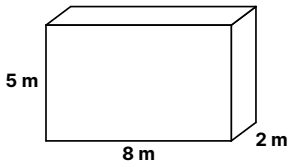
d)



e)



f)



6. Find the volume of:

a) A rectangular prism 15 cm by 4 cm by 2 cm

b) A rectangular prism 30 cm by 10 cm by 5 cm

D Units for length, area and volume ●●○

Remember: A length is measured in units (cm). An area is measured in square units (cm²). A volume is measured in cubic units (cm³).

7. Which unit would you use for each of these: cm, cm² or cm³?

a) The width of a door _____

b) The amount of space inside a drawer _____

c) The size of a phone screen _____

d) The distance around a photo _____

e) The amount of paint that covers a wall _____

f) The amount of water in a fish tank _____

E Problems ●●●

8. A fish tank is 60 cm long, 20 cm wide and 20 cm deep. Find its volume.

9. A box is 5 cm by 2 cm by 2 cm. Find the total volume of 6 of these boxes.

10. A rectangular prism has a volume of 160 cm³. It is 8 cm long and 5 cm wide. Find its height.

11. A cube has a volume of 64 cm³. Find the length of its edges.

★ Extension

12. How many cubes with edges of 2 cm fit exactly into a box that is 8 cm by 4 cm by 2 cm?

13. 36 cubes with edges of 1 cm are used to build a rectangular prism. No edge of the prism is 1 cm long. Write the length, width and height of a prism that could be built.

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